

Problem H1.1 [10 points]

Write a program that gives the roots of the quadratic equation

$$ax^2 + bx + c = 0,$$

where a , b , c are numbers to be defined. To remind you, the roots are given by

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Your program must distinguish between the three cases:

$D < 0$: There are no real solutions

$D = 0$: There is one solution: $x = \dots$

$D > 0$: There are two solutions: $x_1 = \dots$ and $x_2 = \dots$

Make sure that your program takes into account the case where $a = 0$. In this case, give a solution of the linear equation.

Declare the variables a , b and c at the beginning of your code (i.e. give them a value without using the input function).

Use the f-string to print the value of the zeros.

Example input and output:

```
in: a = 1
in: b = 0
in: c = -4
out: There are two solutions: x_1 = 2.0 and x_2 = -2.0
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